

News Headline Sentiment Analyzer

Big Data Fundamentals — Final Project

An end-to-end Big Data pipeline that collects live news headlines, cleans and processes them at scale, applies NLP sentiment analysis, and presents findings in an interactive dashboard.

The Problem

Can we detect whether the news is positive or negative — and does it depend on the topic? This project answers that question by building a fully automated pipeline that fetches real headlines, scores their sentiment using AI, and visualises the results.

Key question: Which news topics carry the most negative sentiment, and which are the most positive?

Architecture



Tools & Technologies

Stage	Tool / Library
Data Acquisition	Python, <code>requests</code> , NewsAPI
Preprocessing	Python, <code>pandas</code>
Exploratory Analysis	<code>matplotlib</code> , <code>seaborn</code> , <code>wordcloud</code>
Big Data Processing	Apache PySpark

Text Analytics (NLP)	VADER SentimentIntensityAnalyzer
Visualisation	Tableau Public
Version Control	GitHub

Key Findings

Dataset: 470 news headlines collected across 5 topics (technology, business, health, science, sports)

Overall Sentiment

Sentiment	Count	Percentage
Positive	195	41.5%
Neutral	176	37.4%
Negative	99	21.1%

→ **More than 4 in 10 headlines are positive** — the news is less negative than most people assume.

Sentiment by Topic

- **Technology** has the highest positive sentiment — driven by AI, innovation, and product launch headlines
- **Science** headlines are the most neutral — factual reporting style
- **Business** and **Health** show a balanced mix
- No topic is majority negative — even the most negative topics are only ~21% negative

Top Sources

Business Insider, TechRadar, Forbes, Financial Post, and The New York Magazine were among the most active sources in this dataset.

Business Value

This pipeline could be deployed by media companies, investors, or PR teams to monitor news sentiment around specific topics or brands in real time.

Dashboard

 **Live Dashboard (Tableau Public):**

https://public.tableau.com/app/profile/bian.andrei2532/viz/ProjectBigData_17803114588630/NewsHeadlineSentimentAnalysis?publish=yes

Project Structure

```
news-sentiment-analysis/  
├── src/  
│   ├── fetch_news.py      # Step 1 — fetch headlines from NewsAPI  
│   ├── preprocess.py      # Step 2 — clean and prepare the data  
│   ├── eda.py             # Step 3 — exploratory data analysis  
│   ├── spark_processing.py # Step 4 — PySpark aggregations  
│   └── sentiment_analysis.py # Step 5 — VADER sentiment scoring  
├── data/  
│   ├── raw/               # Raw CSV from NewsAPI (gitignored)  
│   └── processed/         # Cleaned & scored CSVs (gitignored)  
├── data/charts/           # All generated chart images  
├── requirements.txt  
└── README.md
```

Installation & Usage

```
bash  
  
# 1. Clone the repository  
git clone https://github.com/BianAndrei/news-sentiment-analysis.git  
cd news-sentiment-analysis  
  
# 2. Install dependencies  
pip3 install -r requirements.txt  
  
# 3. Add your NewsAPI key to src/fetch_news.py (line 8)  
  
# 4. Run the pipeline in order  
python3 src/fetch_news.py          # Fetches headlines → data/raw/  
python3 src/preprocess.py          # Cleans data → data/processed/  
python3 src/eda.py                 # Generates charts → data/charts/  
python3 src/spark_processing.py     # PySpark analysis  
python3 src/sentiment_analysis.py  # Sentiment scoring → headlines_with_sentiment.csv
```

Author

Andrei Bian — Big Data Fundamentals, Final Project